Matt | Varghese Mathew

Software Systems Architect

Software Systems Architect with 20 years of experience designing large-scale interoperable platforms. Delivered Epic's HL7 FHIR-based CMS 2027 regulatory implementation, standardizing health plan prior authorization workflows for hundreds of customers. Recognized for transforming fragmented integrations into unified, scalable platforms through deep domain expertise and rigorous system design.

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Madison, WI
linkedin.com/in/matt-varghese
github.com/mattvarghese

WORK EXPERIENCE

Software Systems Architect, Health Plans

Epic - Verona, WI — *MARCH 2021 - PRESENT*

Architect for Epic's HL7 FHIR-based CMS 2027 regulatory platform, defining how health plans implement real-time prior authorization workflows using HL7 CRD, DTR, and PAS standards. Systems adopted by hundreds of Epic customers, transforming authorization turnaround from days to near real-time.

- **Led architecture and delivery** of Epic's payer interoperability platform for CMS 2027, collaborating with internal teams, HL7 Da Vinci workgroups, and external vendors to refine and operationalize evolving standards.
- **Established a unified "source of truth" for prior authorization requirements,** eliminating fragmentation across 12+ product modules and reducing new development lead time by ~80%.
- **Architected standardized data models and execution frameworks** to replace inconsistent one-off implementations, significantly improving maintainability and cross-team scalability.
- **Modernized Medical Necessity guideline integrations (MCG, InterQual)** using SMART on FHIR single sign-on, enabling real-time clinical decision workflows and reducing integration defects by ~60%.
- **Built internal simulation and testing frameworks** to emulate third-party systems, enabling end-to-end workflow validation, proactively eliminating defects, and accelerating partner onboarding.

Senior Software Systems Developer, Clinical Informatics

Epic - Verona, WI — *AUGUST 2010 - MARCH 2021*

Architect for clinical decision support (CDS) integrations, web app integrations, and mobile push notifications. Directed adoption of modern FHIR-based interoperability standards and pioneered Epic's first high-availability cloud service on Azure.

- **Architect of Epic's high-availability mobile push notifications platform,** routing millions of weekly notifications via APNS and GCM. Pioneered deployment as one of Epic's first Azure-native production services, reducing operational costs by over 90%.

SKILLS

Architecture & Infrastructure

- SDLC • Distributed Systems
- Microservices • Azure • CI/CD
- Docker • Virtualization • Hadoop
- Maintainable Architecture

Engineering Stack

- TypeScript • React • Web API
- Node • .NET/C# • C/C++
- Python • Git & SVN • ANSI-M
- SQL & NoSQL • Linux & Windows

Specialized Expertise

- LLM API Integration
- Agentic Workflows
- Network Architecture & Routing

Interoperability & Standards

- HL7 FHIR • SMART on FHIR
- CDS Hooks • CRD/DTR/PAS
- ANSI X12 278 • OAuth2
- JSON/JWT/JWK

COLLABORATION

HL7 FHIR Connectathons

Developed prototypes of HL7 specifications in collaboration with participants from other vendors to improve interoperability.

HL7 Da Vinci Burden Reduction

Represented Epic on public calls steering development of standards to reduce admin burden in Prior Authorization.

- **Led enterprise-wide centralization of 3rd-party web application integrations** to establish standards for security and maintainability. Developed a suite of validation tools for Content Security Policy (CSP), CORS, and Web Messaging APIs.
- **Architected Epic's CDS web services framework,** enabling standards-based clinical decision support across multiple clinical applications using HL7 FHIR and CDS Hooks.
- **Redesigned decision support configuration** to align with clinical mental models, improving usability and enabling efficacy feedback loops for providers.
- **Recognized subject matter expert in Web App Security and RESTful integrations,** providing architectural guidance to multiple teams and onboarding external vendors.

KEY PROJECTS

FHIR-based AI Assistant *2025 Q2*

Integrates LLM APIs with electronic patient health record for agentic use-cases. LLM chat web app answers provider and patient questions using real FHIR data. Can be leveraged for AI driven forms, AI decision support, AI care delivery, etc.

JSON Parser for M/Caché *2020*

Spotlights long-term vision in creating powerful generic reusable tooling. Created RFC 8259 compliant JSON Serializer and Deserializer for ANSI-M/Caché. Over 30 of Epic's database AI integrations leverage this serializer and deserializer.

MIPS Pipeline Simulator *2005, 2025*

github.com/mattvarghese/coconut

Highlights expertise in low-level systems, architecture, parallelism, and performance. 5-Stage pipeline simulation of a MIPS microprocessor, with cache simulation, assembler, and terminal app developed with POSIX APIs, C++, Flex, and Bison in 2005. In 2025, developed a Small-C compiler above the assembler with AI assistance.

PRIOR EXPERIENCE

Research Assistant, Condor Distributed Computing

University of Wisconsin - Madison, WI *SEPT 2008 - MAY 2010*

Virtualized Condor jobs with VMware achieving deterministic runtime environments.

Software Engineer, Enterprise Routing

Cisco Systems - Bangalore, India *JUNE 2006 - JULY 2008*

Platform engineering and configuration tooling for Cisco's enterprise class routers.

EDUCATION

Master of Science, Computer Science

University of Wisconsin - Madison *2008-2010*

GPA: 3.94 / 4.0 | Focus: Systems, Networking, Distributed Computing.

Bachelor of Technology, Computer Science and Engineering

National Institute of Technology - Calicut, India *2002-2006*

CGPA: 9.60 / 10.0 | Department Topper and Gold Medalist.

HL7 CDS Hooks, SMART on FHIR

Contributed to HL7 CDS Hooks and HL7 SMART Web Messaging specifications to enhance workflow integration and security.

Epic App Orchard Conferences

Presented thrice on CDS Hooks and CDS web service integration capabilities to educate and onboard third-party vendors.

Epic Mentoring

Mentored engineers and interns across multiple teams, driving adoption of architectural best practices and development of high-performing teams.

Epic Recruitment

Attracted top talent to Epic by representing Epic at university career fairs and conducting candidate interviews.

EXPLORATORY

Web Based SSH Client *2025 Q3*

Non-traditional solution for real world cost reduction avoiding remote-app licenses for moderated shell / text access to servers.

Proof of concept fully browser-based SSH client for unix servers and Caché databases.

github.com/mattvarghese/WebSSHClient

HTTP Auto-responder *2024 Q4*

Development tooling for Web APIs, keeping traffic within the enterprise or local network.

Configurable endpoint allows simultaneous multi-user testing of RESTful APIs.

github.com/mattvarghese/HttpResponder

EPIC CERTIFICATIONS

Web Tech Camp ⁝ Server Tech Camp ⁝ Clarity Data Model ⁝ EpicCare Inpatient Suite ⁝ Order Transmittal ⁝ Referrals and Authorizations ⁝ Tapestry Health Plan Suite

OTHER

- USHPA Novice (H2) rated hang-glider pilot
- Solo endorsed VFR student pilot
- ASA 101,103/104,105 rated sailor
- Learned Elementary Russian as an adult
- Avid reader, ice-skating, volleyball, XC-skiing, bicycling